Supporting Information

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1 Section 1 - Data generation via High-Throughput Experimentation

1.1 Materials

Standard glovebox techniques were utilized for the handling of air-sensitive reagents. All reagents and starting materials were procured and employed as received, without further purification. Solvents were sourced from sealed anhydrous containers to ensure minimal exposure to atmospheric moisture. Catalysts were obtained from commercial suppliers and stored within an MBraun glovebox, maintaining an inert. 1 mL Clear Glass Shell Vials, 8 x 30 mm (Analytical Sales Services) and 96-position reactor blocks (Analytical Sales Services) were used for HTE in 96 wp. Axygen 384-well Clear V-Bottom 120 µL Polypropylene Deep Well plates were used for HTE in 384wp.

1.2 Instruments

Chronect Quantos were used as solid dispenser. Tecan Fluent, Unchained Lab Junior, SPT Labtech Dragonfly and Apricot PP5 were used as liquid handler for different purpose. IKA MATRIX Orbital Delta+ Shaker was used for shaking and heating the 384 wp in Nano nest (Analytical Sales Services). Ultra-high-performance liquid chromatography-mass spectrometry (UHPLC-MS) analysis was performed using a Waters Acquity system equipped with a SQD2 detector and a Charged Aerosol Detector (CAD).

1.3 Reaction in 96 well plate

The HTE was designed using Katalyst software, which facilitated the generation of automation execution files for both solid dispensing robot and liquid handler robot. All palladium Buchwald precatalysts (1 μmol) and solid bases (12 μmol) were accurately dispensed into 1 mL vials arranged in a 96-well plate, using a Chronect Quantos system within a glovebox environment. Subsequently, 50 μL of aryl halide (ArX) in coresponding solvent stock solution (0.2 M) and 50 μL of amine in coresponding solvent stock solution (0.24 M) were introduced via an Unchained Lab Junior liquid handling system. Finally, the liquid bases (12 μmol) were directly added to the corresponding vials. The resulting reaction mixtures were sealed, stired at 100 °C on Unchained Lab Junior deck within the glovebox for a duration of 16 hours.

1.4 Reaction in 384 well plate (liquid bases only)

The HTE was designed utilizing Katalyst software. A total of 10 µL of 0.2 M aryl halide in dichloroethane (DCE) stock solution and 10 µL of 0.24 M amine in DCE stock solution were dispensed into the corresponding wells of a 384-well plate using a Tecan Fluent liquid handling system. Following this, the plates were subjected to a drying process to remove the solvent using Genevac. Subsequently, the plates were transferred into a glovebox, where 10 µL of palladium Buchwald precatalysts and 10 µL of liquid bases, each dissolved in their respective reaction solvents (0.02 M and 0.24 M, respectively), were added using an SPT Dragonfly liquid dispenser. This ensured that the total volume of solvent for each reaction was 20 µL. After the addition of the reagents, the plates were sealed with Plateloc and placed in a Nano nest. The Nano nest was maintained at a

temperature of 100 °C and agitated at 800 rpm on an orbital shaker for a duration of 16 hours.

1.5 Reaction Analysis

Following the reactions, analytical plates were prepared using an SPT Apricot PP5 system. For the 96-well plates HTE, the reaction mixtures were diluted and mixed with 400 μL of DMSO. Subsequently, 50 μL of the resulting solution was transferred to 450 μL of CHCN for further analysis. For the 384-well plates, the reaction mixtures were diluted and mixed with 40 μL of DMSO. From this solution, 2.5 μL was transferred to 47.5 μL of CHCN. The resulting analytical plates were then centrifuged and analyzed using UPLC-MS/CAD.
Samples for CAD analysis were prepared using a series of noscapine concentrations dissolved in DMSO. The concentrations of the prepared samples were 1.024 mg/mL, 0.512 mg/mL, 0.256 mg/mL, 0.128 mg/mL, 0.064 mg/mL, 0.032 mg/mL, 0.016 mg/mL, and 0.008 mg/mL. Each of these samples, along with the reaction samples, was analyzed using UPLC-MS/CAD, and each sample was run in duplicate.
UPLC Method:

- Solvent A: 0.1
- Solvent B: 0.1
- Gradient: 95
- Stop Time: 2.3 min;
- Flow Rate: 2 mL/min;
- Column: CSH C18, 2.1mm x 50 mm;

Data processing was carried out using Virscidian software. The calibration curve, represented by a quadratic equation, was drawn based on the CAD peak areas of the noscapine calibration samples. The amount of the product in analytical samples was calculated using the CAD peak area corresponding to the products in the experimental samples, in conjunction with the established calibration curve. This process involved incorporating the dilution factors to ascertain the concentration of the product in the original reaction mixture, thereby facilitating an accurate yield calculation.

1.6 Comparative Analysis of 96wp workflow and 384wp workflow

In the validation process, six substrate pairs and 64 reaction conditions were employed. The reactions within the 384 wp format were executed according to the layout depicted below, following the procedure outlined earlier. Subsequently, we compared these results with those obtained from the 96 wp, also following the previously described procedure. To facilitate easier comparison, the results from the 96 wp were reformatted to align with the 384-well layout. In conclusion, out of the 384 reactions, 228 exhibited differences of less than 5

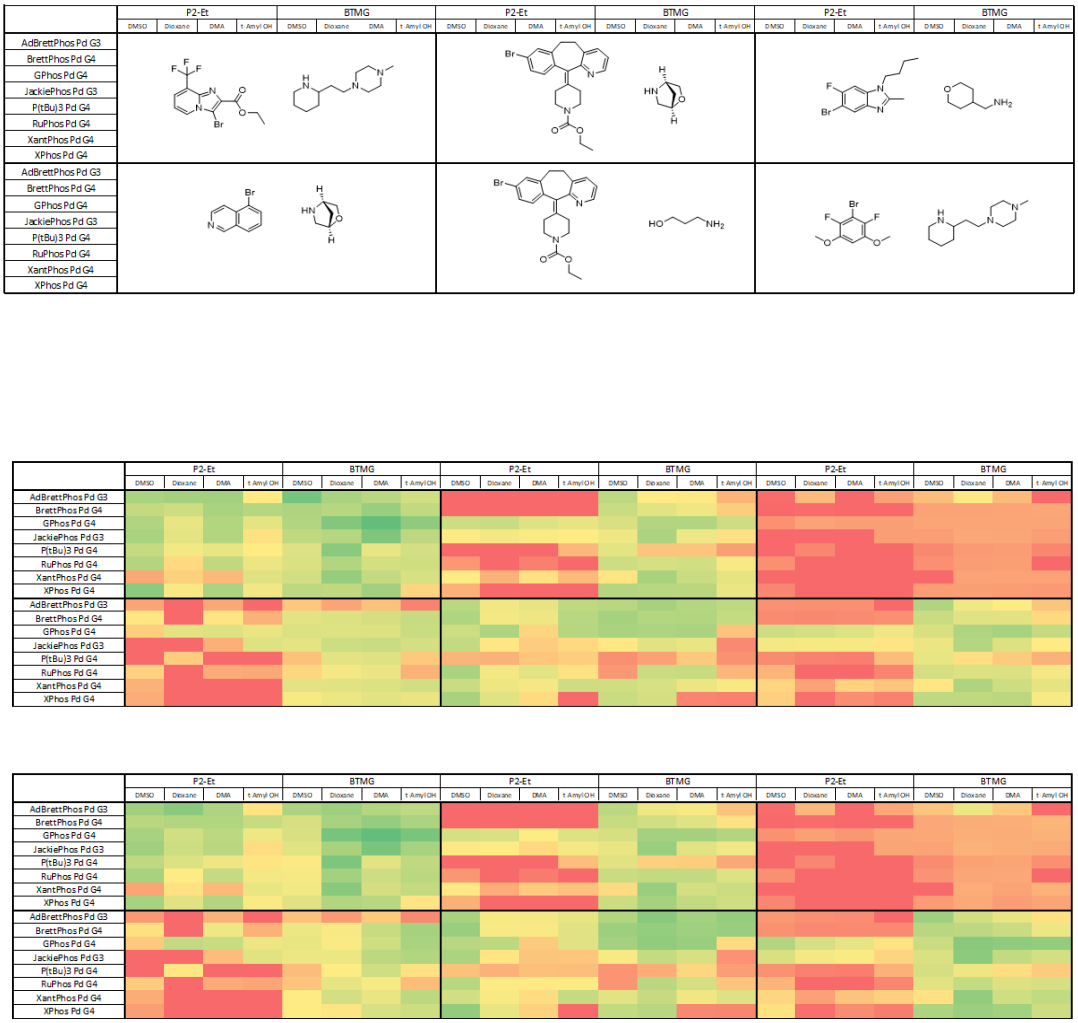

Figure 1: Heatmap of the HTE result from 96 well plate and 384 well plate.

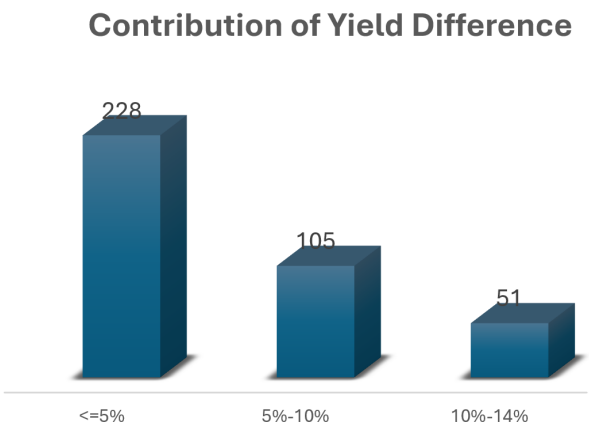

Figure 2: Yield distribution in plate.

2 Section 2 - Selection and Production of Seed Data

The primary objective of this study was to develop a predictive model capable of accurately forecasting Buchwald-Hartwig (BH) reaction outcomes on out-of-distribution (OOD) data. Achieving reliable OOD prediction is essential for synthetic and medicinal chemistry, where diverse reagent combinations are regularly encountered. To meet this goal, we employed a two-stage approach: an initial data production phase to develop a baseline model, followed by a model-guided data production phase, where insights (uncertainty) from the model informed further experimentation to enhance predictive accuracy across broader substrate and reagent spaces.

2.1 Initial Data Production and Baseline Model Development

To build a foundational model capable of accurately predicting Buchwald-Hartwig (BH) reaction outcomes, we began by producing a high-quality, standardized dataset through high-throughput experimentation (HTE). Unlike data typically obtained from electronic lab notebooks (ELNs)—which often lack consistency due to variations in reaction setup, data range, and purification methods—HTE enables standardized reaction conditions and analytical methods. This standardization ensured a high level of reproducibility across a broad set of reaction parameters, including ligand, pre-catalyst, solvent, base, and substrate, all of which significantly impact BH reaction yields. By selecting commercially available Buchwald palladacycle precatalysts, we reduced the reaction space to a manageable scale and allowed for consistency across experiments.

2.2 First Selection and Data Generation

In the first round of data production, we focused on selecting a diverse and representative set of substrates. Specifically focused on aryl bromides, given their ubiquity in our internal work and their relatively high reactivity in cross-coupling.

We started with an initial informer set selection of 76 hand-picked aryl halides, representing scaffold-like structures relevant to pharmaceutical applications. To further define the aryl halide space, we conducted a substructure pattern search, filtering out undesired structures and converting the selected structures to 4096-bit Morgan fingerprints (FPs) at radius 4. This conversion enabled us to quantify structural diversity and check the distribution of the informer set across a combined vendor and proprietary chemical space. After identifying regions lacking representation, we performed random sampling from these uncovered areas and ranked the selected aryl halides by cluster density to ensure adequate coverage across the substrate space.

For the selection of amines, we applied a similar substructure pattern search to define the amine space and removed undesired structures, eg, Sulfonic Acids, Sulfonates, and Sulfonamides, compounds containing both Bromine and Nitrogen or Iodine and Nitrogen which could present issues with reactivity or stability.

The resulting amines were converted into 4096-bit Morgan FPs at radius 3 to assess diversity. Sampling was performed across the entire amine space, and final selections were ranked by cluster density to prioritize amines representative of more densely populated clusters. This approach ensured that selected amines provided broad coverage of the chemical space, supporting the model's ability to generalize across diverse substrates.

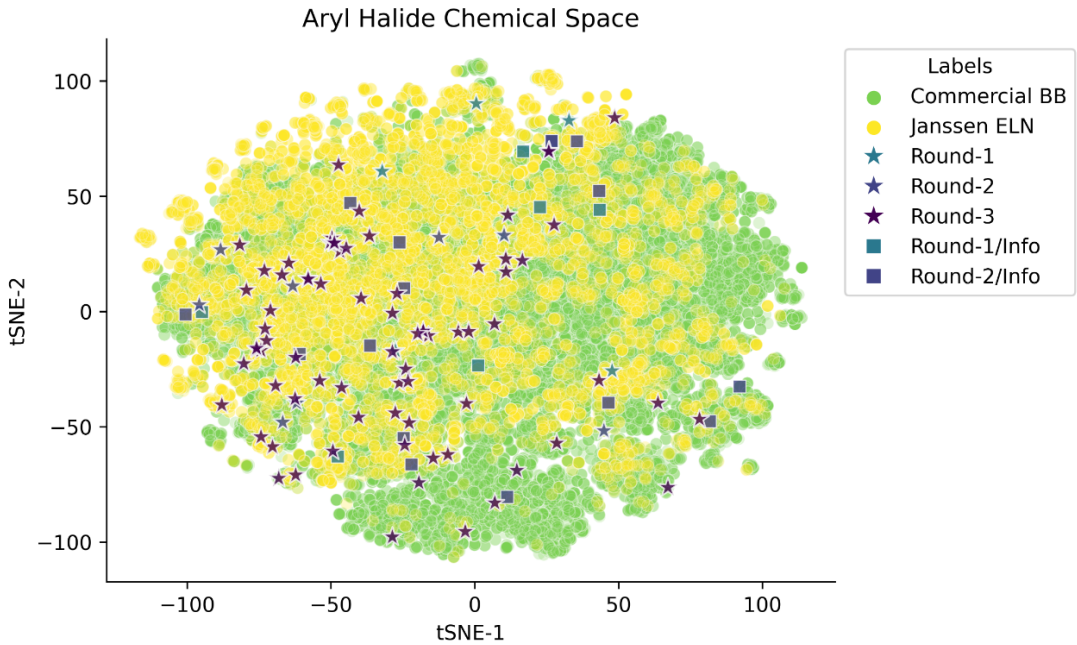

Figure 3: tSNE plot of Aryl Halides selected across different rounds, in commercial and Janssen space.

Additionally, we chose 12 substrate pairs for which the substructures were underrepresented in our ELN data resulting in a set of 2,304 unique BH reactions, incorporating 8 catalysts, 6 bases, and 4 solvents. Product quantification was performed using charged aerosol detection (CAD) to accurately measure yields across various products. This initial round of data collection, although systematically diverse, resulted in low predictive power, indicating a need for a broader substrate scope.

2.3 Expanded Reaction and Substrate Diversity

To improve the performance and scope of the seed model, a second round of data collection expanded both the substrate diversity and reaction conditions. We selected a matrix of 24 aryl halides and 24 amines, incorporating 32 Buchwald palladacycle precatalysts, 6 bases, and 4 solvents, ultimately producing 3,840 reactions. Each pair of coupling partners was tested under multiple conditions (4 to 8 randomly selected combinations of catalysts, bases, and solvents) to capture a wider spectrum of possible outcomes.

The experimental workflow was optimized for automation, using polypropylene 384-well plates at a 1 µmol scale to facilitate high-throughput reaction setup and analysis. Automated liquid handling robots (SPT Dragonfly) were employed to streamline this process, enabling the completion of reactions within approximately 20 days. This expanded dataset significantly enhanced the model's predictive performance, and uncertainty estimation, providing greater diversity in reaction outcomes and substrate interactions, which proved essential for achieving reliable OOD predictions.

The first stage of data production thus established a robust and diverse foundation for our BH model by systematically expanding the substrate and reaction conditions. Through carefully selected rounds of HTE-driven data collection, this stage laid the groundwork for a model capable of generalizing across novel reagent combinations, setting the stage for subsequent refinement and model-guided data production in Stage 2.

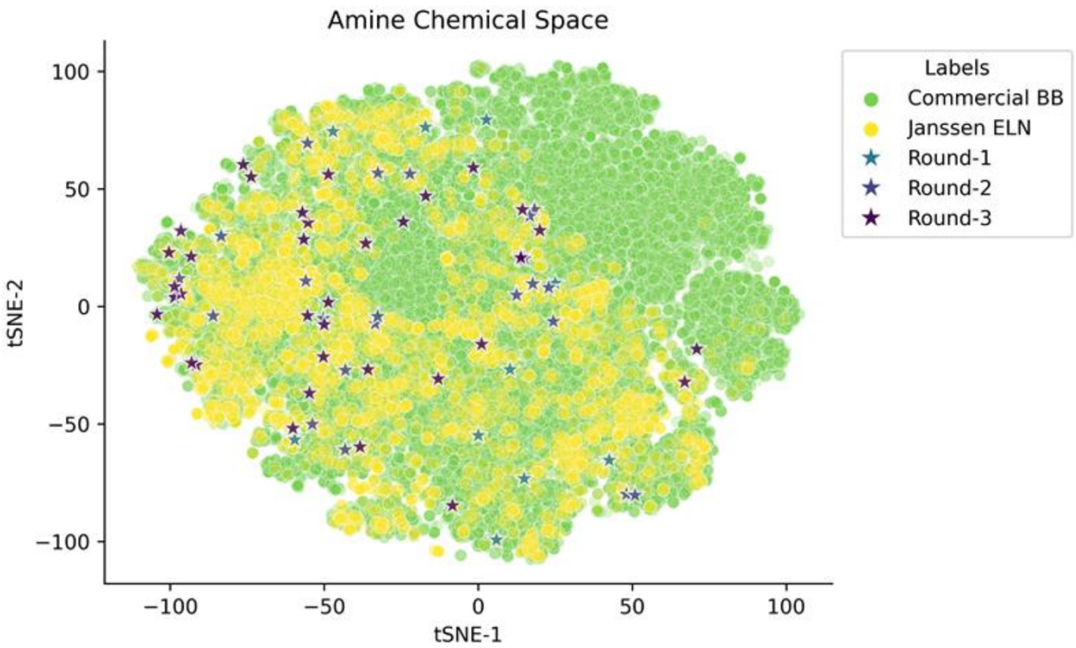

Figure 4: tSNE plot of Amines selected across different rounds, in commercial and Janssen space.

3 Section 3 - Model-Guided Data Production

3.1 Overview

Stage 2 operationalized active learning over a virtual Buchwald-Hartwig (BH) reaction space with 65.6M candidate reactions using a BERT-Enriched Embeddings (BEE) success estimation model fine-tuned on Stage 1 seed data. New experiments were selected in random batch mode active learning (RBMAL) with an acquisition function that (i) targets model uncertainty for specific, fully specified reactions (ArX, amine, Pd precatalyst/ligand, base, solvent), (ii) enforces diversity via Tanimoto distance from historical + in-batch selections, and (iii) respects substrate-space density and pharmaceutical applicability constraints. Substrate choices followed Stage 1 criteria (diversity + density + applicability); RBMAL prioritized reagents/conditions for those substrates.

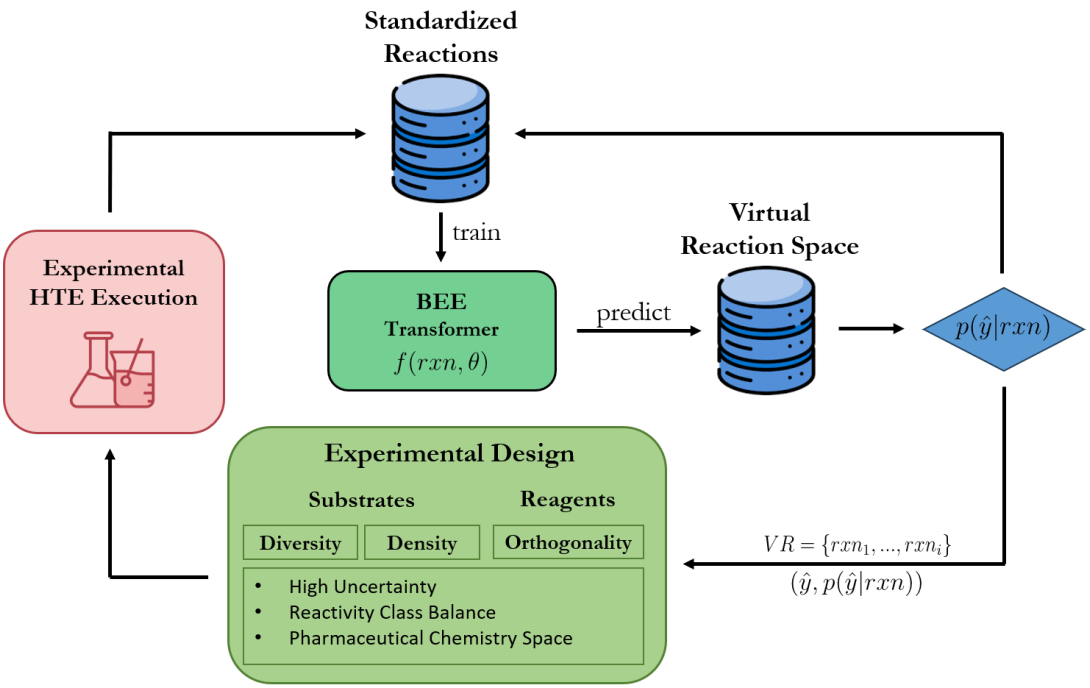

Figure 5: Diagram with active learning flow.

3.2 Virtual BH Reaction Space

We enumerated a constrained, pharmaceutically relevant BH space by Cartesian product of curated dictionaries:

- Aryl halides (Ar-X): N = 186
- Amines (NuH): N = 57
- Pd sources/ligand systems: N = 43 Buchwald palladacycles: AdBrettPhos Pd G3, BrettPhos Pd G4, GPhos Pd G4, JackiePhos Pd G3, P(tBu)3 Pd G4, RuPhos Pd G4, XantPhos Pd G4, XPhos Pd G4, APhos Pd G4, cataCXium Pd G4, CPhos Pd G3, CyJohnPhos Pd G4, DavePhos Pd G3, DPPF Pd G4, DTBPF Pd G3, EPhos Pd G4, JosiPhos SL-J009-1 Pd G3, Me4tBuXPhos Pd G3, MorDalPhos Pd G4, N-XantPhos Pd G4, (o-tol)3 Pd G2, PCy3 Pd G4, QPhos Pd G3, rac-BINAP Pd G4,

Pd G3, SPhos Pd G2, SPhos Pd G4, Pd G3, tBuDavePhos Pd G4, tBuXPhos Pd G3, TrixiePhos Pd G3, XPhos Pd G3, MeCgPPh Pd G3, Neopentyl(t-Bu)2P Pd G4, (t-Bu)2PhP Pd G4, Me3(OMe)tBuXPhos Pd G3, (t-Bu)PhCPhos Pd G4, VPhos Pd G4, (tBu)2PMe Pd G4, (R)-TolBINAP Pd G4, EtCPhos Pd G3, NJackiePhis Pd G4, (tBu)2P(nBu) Pd G4

- Bases: N = 12: P2-Et, Cs2CO3, NaOtBu, LiHMDS, NaBHT, BTMG, NaOTMS, DBU, KOPiv, K3PO4, MTBD, Cy2NMe
- Solvents: N = 12: t-Amyl OH, DMSO, DMA, 1,4-Dioxane, Toluene, MeCN, 2-Me-THF, DMF, CPME, NMP, iPrOAc, DME

Substrate selection followed Stage 1 logic:
cluster substrates (aryl ring system for halides; substructure for amines),
filter clusters absent from ELN (to ensure translational relevance),
rank candidates by cluster density (coverage proxy) and pharmaceutical applicability (availability/IP/safety),
pick cluster representatives with varying complexity.
After structural/availability/toxicity filters, enumeration yielded 10,602 potential products and 65.6M fully specified reactions (product, condition tuple). To account for diversity, we clustered the substrates represented with Morgan (ECFP) fingerprints (Ar-X: radius 4, 4096 bits; amines: radius 3, 4096 bits), as in Stage 1.
Once the substrate panel for an iteration was fixed, RBMAL optimized reagents/conditions.

3.3 RBMAL acquisition function and batch construction

Ranked batch-mode active learning (RBMAL) is an acquisition function that combines a model's uncertainty and a distance metric into a single score, as defined in Eq. 1. The score is calculated as:

$$\text{Score} = \alpha \cdot (1 - \text{Similarity}) + (1 - \alpha) \cdot \text{Uncertainty} \tag{1}$$

The parameter α is given by:

$$\alpha = \frac{|U|}{|U| + |D|} \tag{2}$$

where U and D represent the unlabeled and labeled sets, respectively.
In our study, we used the Tanimoto distance computed from Morgan fingerprints as the similarity measure. To select substrates, all 186 aryl halides were first ranked using Eq. 1, based on the mean confidence from 65 million predicted reactions. The top 75 aryl halides were then selected. For each aryl halide, we recursively selected 4 unique amines from a short-listed set of 57 amines, and 16 conditions for each pair, again utilizing Eq. 1 at both selection steps to ensure maximum diversity and informativeness.
The final experimental set included the 75 selected aryl halides, 37 amines, and a total of 714 combinations of catalyst, base, and solvent conditions.

3.4 Iterative loop and retraining schedule

Each active learning iteration followed this sequence:

- Freeze Transformer (BEE) trained on Stage 1 + all prior Stage 2 data.
- Score the virtual space restricted to the current substrate panel; build RBMAL batches as described in S3.3.
- Execute HTE (384-well or 96-well plates): automated setup, CAD/UPLC-MS readouts, standardized processing.
- Ingest data to tabular format: structure standardization, add role columns, canonicalize smiles and prepare into BEE compatible format.
- Retrain/fine-tune BEE with early stopping and validate on OOD substrates.

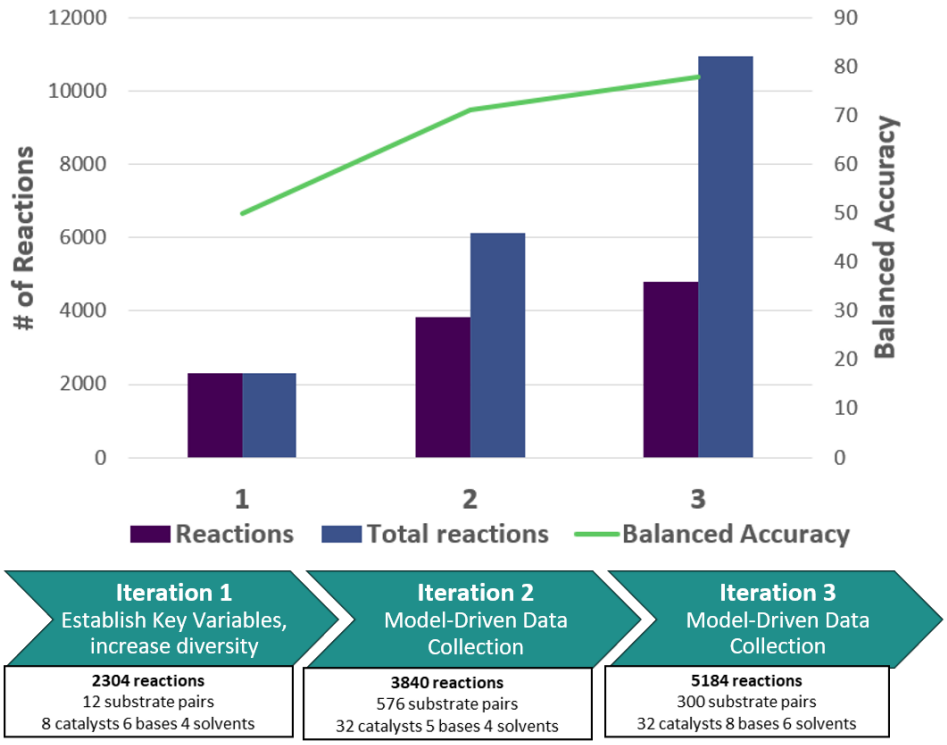

Figure 6: Bar plot with number of reactions executed per iteration, and average balance accuracy across 20 train-test splits where substrates in test are not present in the train split.

In iteration2, the first model driven data collect, we significantly increased catalyst coverage and produced 3840 reaction samples. The resulting model was validated by creating 20 train-test splits where any substrate selected for the test were removed from the train, while the conditions are within domain for the model. The result of iteration 2 showed the first measure of generalizability for novel substrates with a balanced accuracy of 72%. The following iteration we maintained the reagent scope with a small increase in base and solvents and produced our largest batch of 5184 reactions. The resulting model showed an improvement in performance when predicting novel substrates with a balanced accuracy of 80%.
It's worth noting that despite having an ambitious reagent space coverage not all reagents were equally screened, with 9 of the 32 catalysts having a sample count under 100. Due

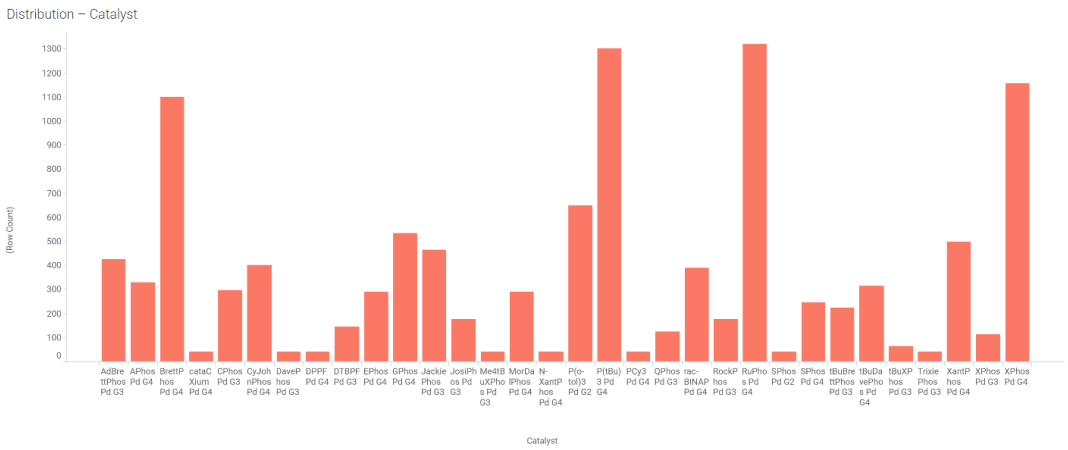

Figure 7: Catalysts used and respective sample count.

to the nature of the OOD validation already described, these catalysts are represented according to the distribution in the train set and have low impact on performance metrics. Regarding the bases and solvents, all with exception of NaOTMS have a sample count above 500 and are well within domain of the covered reagent space.

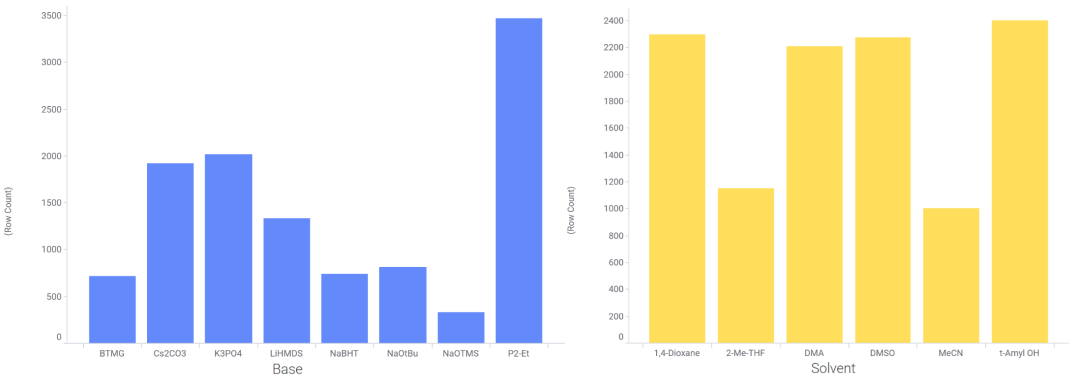

Figure 8: Sample count of bases and solvents used.

4 Section 4 - Combining BH HTE Literature and Benchmarking

4.1 Compound–Reaction Diversity Score (CRDS)

To quantitatively assess the diversity of chemical datasets, we developed the **Compound–Reaction Diversity Score (CRDS)**. This metric integrates contributions from three key descriptors: (i) the number of distinct reactants (R_t), (ii) the number of distinct reagents (R_g), and (iii) the number of distinct reactions (R_n). Each descriptor is normalized against the maximum observed value within the collection and is weighted according to tunable exponents that allow dataset-specific prioritization.
The general form of CRDS is given by:

$$\text{CRDS} = \left(\frac{R_t}{R_t^{\max}}\right)^{\alpha} \times \left(\frac{\min(R_g, T_g)}{R_g^{\max}}\right)^{\beta_{\mathrm{u}}} \times \left(\max\left(\frac{R_g}{T_g}, 1\right)\right)^{\beta_{\mathrm{d}}} \times \left(\frac{\min(R_n, T_n)}{R_n^{\max}}\right)^{\gamma_{\mathrm{u}}} \times \left(\max\left(\frac{R_n}{T_n}, 1\right)\right)^{\gamma_{\mathrm{d}}}$$

Here,

- $R_t^{\max}, R_g^{\max}, R_n^{\max}$ denote the maximum values of reactants, reagents, and reactions across all datasets, ensuring normalization.
- $\alpha, \beta_{\mathrm{u}}, \beta_{\mathrm{d}}, \gamma_{\mathrm{u}}, \gamma_{\mathrm{d}}$ are weighting exponents that allow differential emphasis on each descriptor.
- T_g and T_n are *saturation thresholds*, beyond which additional diversity yields diminishing returns.

The structure of CRDS ensures that both breadth (coverage of reactants, reagents, and reactions up to the threshold values) and depth (expansion beyond the thresholds) are explicitly represented:

- For R_g and R_n, the "up" term ($\min(\cdot)$) emphasizes dataset growth up to the saturation threshold (T_g, T_n), rewarding balanced coverage.
- The "down" term ($\max(\cdot)$) introduces a diminishing-returns regime for values exceeding the threshold, preventing overinflation of the score by datasets dominated by excessive reagents or reactions without balanced representation elsewhere.

This design provides a nonlinear scoring scheme that prioritizes diverse yet balanced datasets, reflecting both practical chemical intuition (favoring expansion up to an optimal level) and statistical robustness (avoiding domination by single descriptors). The weighting exponents may be tuned to emphasize different chemical contexts—for example, increasing β_{up} to prioritize reagent diversity in synthetic methodology datasets, or increasing γ_{up} to emphasize reaction coverage in retrosynthetic benchmarks.

4.2 Comparison with Simplified CRDS and Sample Size

To test whether CRDS provides additional explanatory power compared to dataset size alone, we defined a **simplified CRDS** as:

$$CRDS_{\text{simple}} = \left(\frac{R_t}{R_t^{\max}}\right)^{\alpha} \times \left(\frac{R_g}{R_g^{\max}}\right)^{\beta} \times \left(\frac{R_n}{R_n^{\max}}\right)^{\gamma}$$

This form omits the saturation thresholds and diminishing-returns terms, reducing the score to a straightforward normalized product of reactant, reagent, and reaction counts. To quantify the relationship between dataset characteristics and out-of-domain model performance, we computed Pearson and Spearman correlation coefficients between predictive accuracy and (i) dataset sample size, (ii) simplified CRDS, and (iii) full CRDS. The results are summarized in Table 1.

Table 1: Performance Metrics - Pearson and Spearman Correlations

	Pearson				**Spearman**			
	ROC AUC	**Bal. Acc.**	**F1 Score**	**AU-PR-C**	**ROC AUC**	**Bal. Acc.**	**F1 Score**	**AU-PR-C**
CRDS	**0.897**	0.814	0.409	**0.889**	**0.967**	**0.900**	0.517	**0.967**
Simplified CRDS	0.884	**0.820**	**0.449**	0.865	0.933	0.833	**0.600**	**0.967**
Dataset Size	0.588	0.614	0.193	0.667	0.650	0.700	0.117	0.617

These results demonstrate that while dataset size correlates weakly with out-of-domain performance, incorporating diversity-aware descriptors via CRDS produces a substantially stronger association. This underscores the value of CRDS as a principled metric for benchmarking dataset utility in chemical machine learning.

S4.2 Comparative value of literature HTE sources for OOD prediction of BH outcomes

There were two goals to be addressed in the in-silico validation portion of this project:

- To quantify how informative each individual open-source BH HTE dataset is for training models that generalize to novel substrates.
- To benchmark All w/o JnJ24 (all literature sources combined, excluding our HTE data) and All w JnJ24 (all literature + our HTE data).

Towards that end we trained multiple ML models per training corpus and evaluated them on outer clusters that are sparsely represented by the training corpus, over multiple random cluster initializations and calculate Mean ± std per metric. To prepare the data we operated on a pre-curated table in which chemical entities were normalized across sources (beyond canonicalization), ensuring that the same reagent has a single, unified representation in the final dataset. This greatly impacts ability to learn from multiple data sources. Each row encodes the source of the sample (Source column) and a single reaction with its with unified entity columns (e.g., Aryl Halide, Amine, Pd system, Base, Solvent) which inherently include role information. Role is an interesting variable which allows different encoding strategies of the chemical entity depending role, an opportunity for model improvement in future work.
It's worth noting that this cross-source normalization step is only relevant when combining sources, during the acquisition stage of the active learning cycle we did not rely on this harmonization.

- **Reactivity class**: The models trained were exclusively binary classification models, and we defined the positive class as reactions that yielded over 10% producing $labels \in \{0, 1\}$.
- **Features**: Due to the large number of training sets we wanted to test (9), and the combinatorial increase with 20 random cluster initializations we opted for Reaction Fingerprints, DRFP with $radius = 3$ and $nBits = 2048$. For this particular benchmark it was not practical to train-test large models like BEE hundreds or thousands of times ($9 \times 20 \times N$ models $\times$ N experiments).
- **Clustering and OOD evaluation protocol**: We used k-means clustering in reaction/feature space to create structure-aware outer-cluster tests, with K = 12 clusters. For each training corpus S (a single literature source) we count the number of training examples per cluster and define outer clusters as those where S contributes 30 examples. The evaluation set is all reactions (across sources) whose cluster is in this outer set (and removing any examples from S). This creates an approximate OOD regime: predictions are scored on clusters that are poorly covered by the training corpus, reducing in-domain leakage.

4.3 Models and Training

For each corpus S and run, we train six classifiers on DRFP features using a common training routine (scalers, where required, are fit only on the training data):

- Random Forest (`rf_model`)
- Gradient Boosting (`gradient_boosting_model`)
- XGBoost (`xgb_model`)
- Logistic Regression (`lr_model`)
- Multi-Layer Perceptron (`mlp_model`)
- k-Nearest Neighbors (`knn_model`)

Training corpora S: (i) each literature `Source` individually; (ii) **(All w/o JnJ24)**: union of all literature sources excluding our HTE data; (iii) **All w JnJ24**: union of all literature sources plus our HTE data. The same clustering protocol applies in all cases in (i), and outer clusters are defined relative to S. For (ii) and (iii) the we select 8 cluster for the train set and 4 clusters to the test set.

To evaluate the models and reduce the possible bias introduced by any given metric we calculated performance on the outer-cluster evaluation set, across multiple metrics:

- Balanced Accuracy (BA)
- ROC AUC
- F1 score (positive class = 1)
- F0 score (positive class = 0)

Model	**Hyperparameter**	**Value**
Random Forest	Number of estimators (`n_estimators`)	300
	Maximum depth (`max_depth`)	None
	Minimum samples split (`min_samples_split`)	4
	Minimum samples leaf (`min_samples_leaf`)	3
	Splitting criterion (`criterion`)	Entropy
Gradient Boosting	Number of estimators (`n_estimators`)	100
XGBoost	Evaluation metric (`eval_metric`)	Logloss
Logistic Regression	Maximum iterations (`max_iter`)	400
Multi-Layer Perceptron (MLP)	Hidden layer sizes (`hidden_layer_sizes`)	(100, 20)
	Solver (`solver`)	L-BFGS
k-Nearest Neighbors	Number of neighbors (`n_neighbors`)	5

Table 2: Hyperparameter settings for models.

- AU-PR-C (area under precision–recall curve for class 1)
- Precision (class 1)
- Recall (class 1)

Binary thresholding affects class balance, therefore we report both threshold-insensitive metrics (ROC-AUC, AU-PR-C) and threshold-dependent ones (F1, BA). In order to tackle stochastic bias, we perform 20 independent runs (`run = 0..20`), each with a new k-means `random_state=run`. For each (S, model) we train on all rows from S (shuffled with `random_state_key=run`), evaluate on that run's outer clusters, and append metrics. The summary Table 3 contains the average and standard deviation for each metric, for the best performing model in each train source. For the full results table view Table 4.

Table 3: Best Performance OOD Clusters (DRFP)

Model	train$_data$	test$_data$	**Balanced Accuracy**		**ROC AUC**		**F1 Score**		**F0 Score**		**AU-PR-C**		**Precision 1**		**Recall 1**	
Model	Train Data	Test Data	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std
gaussianNB_model	AZ ELN 2023	Outter Clusters	0.5	0.0	0.5	0.0	0.63	0.01	0.01	0.01	0.73	0.0	0.46	0.01	0.99	0.01
rf_model	All With JnJ	Outter Clusters	0.75	0.02	0.84	0.02	0.67	0.04	0.82	0.01	0.81	0.03	0.83	0.02	0.59	0.04
knn_model	All without JnJ	Outter Clusters	0.62	0.03	0.67	0.04	0.56	0.04	0.64	0.03	0.63	0.04	0.56	0.04	0.6	0.06
xgb_model	Dreher 2014	Outter Clusters	0.5	0.0	0.61	0.01	0.0	0.0	0.74	0.01	0.48	0.01	0.0	0.0	0.0	0.0
xgb_model	Dreher 2016	Outter Clusters	0.57	0.01	0.56	0.01	0.6	0.01	0.52	0.03	0.51	0.01	0.54	0.01	0.69	0.03
xgb_model	Dreher Doyle 2018	Outter Clusters	0.5	0.0	0.48	0.0	0.55	0.0	0.09	0.0	0.38	0.0	0.38	0.0	0.95	0.0
lr_model	HiTEA 2024	Outter Clusters	0.59	0.01	0.6	0.01	0.61	0.02	0.57	0.03	0.58	0.03	0.59	0.03	0.63	0.01
knn_model	JNJ HTE 2024	Outter Clusters	0.6	0.02	0.61	0.02	0.65	0.03	0.53	0.06	0.71	0.02	0.58	0.02	0.73	0.05
gradient_boosting_model	Roche 2023	Outter Clusters	0.59	0.01	0.6	0.01	0.51	0.01	0.66	0.01	0.58	0.02	0.58	0.02	0.45	0.01

Table 4: All Performance Results (DRFP)

Model	Features	Train Data	Test Data	**Balanced Accuracy**		**ROC AUC**		**F1 Score**		**AU-PR-C**		**F0 Score**		**Precision 1**		**Recall 1**	
Model	Features	Train Data	Test Data	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std
rf_model	DRFP	All With JnJ	Outter Clusters	0.75	0.02	0.84	0.02	0.67	0.04	0.81	0.03	0.82	0.01	0.83	0.02	0.59	0.04
xgb_model	DRFP	All With JnJ	Outter Clusters	0.73	0.03	0.82	0.02	0.65	0.06	0.79	0.03	0.79	0.02	0.78	0.03	0.58	0.08
lr_model	DRFP	All With JnJ	Outter Clusters	0.67	0.03	0.74	0.03	0.61	0.05	0.68	0.04	0.71	0.03	0.64	0.04	0.6	0.09
mlp_model	DRFP	All With JnJ	Outter Clusters	0.71	0.02	0.79	0.01	0.65	0.03	0.75	0.02	0.76	0.02	0.7	0.03	0.62	0.06
gradient_boosting_model	DRFP	All without JnJ	Outter Clusters	0.63	0.02	0.67	0.03	0.5	0.04	0.66	0.03	0.73	0.01	0.67	0.02	0.42	0.04
xgb_model	DRFP	All without JnJ	Outter Clusters	0.6	0.03	0.64	0.04	0.43	0.08	0.61	0.05	0.72	0.02	0.66	0.05	0.34	0.07
gaussianNB_model	DRFP	All without JnJ	Outter Clusters	0.49	0.03	0.49	0.03	0.45	0.05	0.59	0.04	0.48	0.04	0.43	0.05	0.51	0.09
lr_model	DRFP	All without JnJ	Outter Clusters	0.52	0.03	0.55	0.03	0.46	0.06	0.46	0.03	0.57	0.03	0.46	0.03	0.48	0.08
mlp_model	DRFP	All without JnJ	Outter Clusters	0.56	0.03	0.58	0.02	0.47	0.07	0.54	0.02	0.62	0.03	0.51	0.03	0.45	0.1
knn_model	DRFP	All without JnJ	Outter Clusters	0.62	0.03	0.67	0.04	0.56	0.04	0.63	0.04	0.64	0.03	0.56	0.04	0.6	0.06
rf_model	DRFP	All without JnJ	Outter Clusters	0.64	0.03	0.69	0.03	0.47	0.09	0.67	0.03	0.75	0.01	0.76	0.03	0.37	0.08
gradient_boosting_model	DRFP	Dreher 2014	Outter Clusters	0.5	0.0	0.52	0.03	0.0	0.0	0.42	0.02	0.74	0.01	0.43	0.3	0.0	0.0
xgb_model	DRFP	Dreher 2014	Outter Clusters	0.5	0.0	0.61	0.01	0.0	0.0	0.48	0.01	0.74	0.01	0.0	0.0	0.0	0.0
gaussianNB_model	DRFP	Dreher 2014	Outter Clusters	0.5	0.0	0.5	0.0	0.0	0.0	0.7	0.01	0.74	0.01	0.0	0.0	0.0	0.0
lr_model	DRFP	Dreher 2014	Outter Clusters	0.5	0.0	0.5	0.01	0.0	0.0	0.43	0.01	0.74	0.01	0.0	0.0	0.0	0.0
mlp_model	DRFP	Dreher 2014	Outter Clusters	0.5	0.0	0.44	0.02	0.01	0.0	0.38	0.01	0.74	0.01	0.5	0.02	0.0	0.0
knn_model	DRFP	Dreher 2014	Outter Clusters	0.51	0.0	0.44	0.02	0.12	0.02	0.38	0.02	0.73	0.02	0.45	0.03	0.07	0.01
gradient_boosting_model	DRFP	Dreher 2016	Outter Clusters	0.49	0.0	0.57	0.01	0.14	0.02	0.51	0.02	0.66	0.02	0.42	0.05	0.08	0.01
xgb_model	DRFP	Dreher 2016	Outter Clusters	0.57	0.01	0.56	0.01	0.6	0.01	0.51	0.01	0.52	0.03	0.54	0.01	0.69	0.03
gaussianNB_model	DRFP	Dreher 2016	Outter Clusters	0.5	0.0	0.5	0.0	0.05	0.0	0.51	0.02	0.68	0.02	0.53	0.02	0.03	0.0
lr_model	DRFP	Dreher 2016	Outter Clusters	0.49	0.0	0.5	0.01	0.1	0.01	0.47	0.02	0.67	0.02	0.41	0.03	0.06	0.0
mlp_model	DRFP	Dreher 2016	Outter Clusters	0.52	0.01	0.52	0.01	0.27	0.01	0.52	0.02	0.66	0.02	0.54	0.02	0.18	0.01
knn_model	DRFP	Dreher 2016	Outter Clusters	0.51	0.0	0.55	0.01	0.09	0.0	0.53	0.01	0.69	0.02	0.64	0.01	0.05	0.0
gradient_boosting_model	DRFP	Dreher Doyle 2018	Outter Clusters	0.51	0.0	0.44	0.01	0.55	0.0	0.34	0.01	0.15	0.0	0.39	0.0	0.93	0.0
xgb_model	DRFP	Dreher Doyle 2018	Outter Clusters	0.5	0.0	0.48	0.0	0.55	0.0	0.38	0.0	0.09	0.0	0.38	0.0	0.95	0.0
gaussianNB_model	DRFP	Dreher Doyle 2018	Outter Clusters	0.49	0.0	0.48	0.0	0.51	0.0	0.56	0.0	0.29	0.0	0.38	0.0	0.78	0.0
lr_model	DRFP	Dreher Doyle 2018	Outter Clusters	0.5	0.0	0.48	0.0	0.55	0.0	0.37	0.0	0.07	0.0	0.38	0.0	0.95	0.0
mlp_model	DRFP	Dreher Doyle 2018	Outter Clusters	0.49	0.0	0.46	0.0	0.55	0.0	0.35	0.0	0.05	0.0	0.38	0.0	0.96	0.0
knn_model	DRFP	Dreher Doyle 2018	Outter Clusters	0.49	0.0	0.48	0.01	0.52	0.0	0.47	0.01	0.27	0.0	0.38	0.0	0.82	0.01
gradient_boosting_model	DRFP	HiTEA 2024	Outter Clusters	0.52	0.0	0.56	0.01	0.26	0.01	0.56	0.04	0.64	0.03	0.57	0.05	0.17	0.0
xgb_model	DRFP	HiTEA 2024	Outter Clusters	0.5	0.0	0.47	0.01	0.03	0.0	0.48	0.04	0.65	0.03	0.43	0.07	0.02	0.0
gaussianNB_model	DRFP	HiTEA 2024	Outter Clusters	0.53	0.0	0.53	0.0	0.63	0.03	0.71	0.03	0.39	0.02	0.53	0.04	0.77	0.01
lr_model	DRFP	HiTEA 2024	Outter Clusters	0.59	0.01	0.6	0.01	0.61	0.02	0.58	0.03	0.57	0.03	0.59	0.03	0.63	0.01
mlp_model	DRFP	HiTEA 2024	Outter Clusters	0.52	0.01	0.55	0.01	0.65	0.03	0.7	0.03	0.27	0.03	0.52	0.04	0.87	0.01
knn_model	DRFP	HiTEA 2024	Outter Clusters	0.49	0.01	0.4	0.01	0.02	0.01	0.4	0.04	0.64	0.03	0.26	0.07	0.01	0.0
gradient_boosting_model	DRFP	AZ ELN 2023	Outter Clusters	0.54	0.01	0.56	0.01	0.56	0.02	0.49	0.01	0.5	0.02	0.49	0.01	0.65	0.03
xgb_model	DRFP	AZ ELN 2023	Outter Clusters	0.53	0.0	0.54	0.0	0.58	0.0	0.5	0.01	0.45	0.01	0.49	0.01	0.71	0.0
gaussianNB_model	DRFP	AZ ELN 2023	Outter Clusters	0.5	0.0	0.5	0.0	0.63	0.01	0.73	0.0	0.01	0.01	0.46	0.01	0.99	0.01
lr_model	DRFP	AZ ELN 2023	Outter Clusters	0.42	0.01	0.41	0.01	0.41	0.01	0.4	0.01	0.44	0.0	0.39	0.01	0.43	0.01
mlp_model	DRFP	AZ ELN 2023	Outter Clusters	0.47	0.01	0.35	0.01	0.11	0.01	0.39	0.01	0.65	0.0	0.32	0.03	0.07	0.01
knn_model	DRFP	AZ ELN 2023	Outter Clusters	0.48	0.0	0.42	0.01	0.6	0.01	0.48	0.01	0.12	0.01	0.45	0.01	0.9	0.0
gradient_boosting_model	DRFP	Roche 2023	Outter Clusters	0.59	0.01	0.6	0.01	0.51	0.01	0.58	0.02	0.66	0.01	0.58	0.02	0.45	0.01
xgb_model	DRFP	Roche 2023	Outter Clusters	0.5	0.0	0.52	0.01	0.09	0.01	0.47	0.02	0.69	0.01	0.43	0.03	0.05	0.0
gaussianNB_model	DRFP	Roche 2023	Outter Clusters	0.49	0.0	0.49	0.0	0.04	0.02	0.38	0.02	0.69	0.01	0.3	0.07	0.02	0.01
lr_model	DRFP	Roche 2023	Outter Clusters	0.48	0.01	0.46	0.01	0.39	0.0	0.42	0.02	0.56	0.01	0.43	0.03	0.36	0.01
mlp_model	DRFP	Roche 2023	Outter Clusters	0.55	0.01	0.52	0.01	0.48	0.01	0.55	0.02	0.62	0.01	0.52	0.03	0.45	0.01
knn_model	DRFP	Roche 2023	Outter Clusters	0.49	0.0	0.51	0.01	0.56	0.01	0.57	0.02	0.33	0.01	0.45	0.02	0.75	0.0
gradient_boosting_model	DRFP	JNJ HTE 2024	Outter Clusters	0.53	0.01	0.57	0.01	0.38	0.03	0.54	0.04	0.62	0.02	0.56	0.04	0.29	0.02
xgb_model	DRFP	JNJ HTE 2024	Outter Clusters	0.5	0.01	0.51	0.01	0.35	0.02	0.5	0.04	0.59	0.02	0.5	0.04	0.27	0.02
gaussianNB_model	DRFP	JNJ HTE 2024	Outter Clusters	0.53	0.0	0.53	0.0	0.59	0.02	0.68	0.02	0.45	0.02	0.52	0.03	0.66	0.02
lr_model	DRFP	JNJ HTE 2024	Outter Clusters	0.49	0.01	0.5	0.01	0.46	0.0	0.49	0.03	0.52	0.03	0.49	0.02	0.43	0.01
mlp_model	DRFP	JNJ HTE 2024	Outter Clusters	0.55	0.01	0.57	0.02	0.58	0.02	0.64	0.02	0.52	0.04	0.55	0.02	0.61	0.01
knn_model	DRFP	JNJ HTE 2024	Outter Clusters	0.6	0.02	0.61	0.02	0.65	0.03	0.71	0.02	0.53	0.06	0.58	0.02	0.73	0.05
gradient_boosting_model	DRFP	JNJ HTE+PMC 2024	Outter Clusters	0.61	0.0	0.67	0.01	0.74	0.0	0.75	0.0	0.49	0.01	0.71	0.0	0.78	0.0
xgb_model	DRFP	JNJ HTE+PMC 2024	Outter Clusters	0.57	0.0	0.59	0.0	0.72	0.0	0.69	0.01	0.43	0.0	0.68	0.01	0.76	0.0
gaussianNB_model	DRFP	JNJ HTE+PMC 2024	Outter Clusters	0.5	0.0	0.5	0.0	0.78	0.0	0.82	0.0	0.0	0.0	0.63	0.01	1.0	0.0
lr_model	DRFP	JNJ HTE+PMC 2024	Outter Clusters	0.48	0.0	0.47	0.0	0.54	0.0	0.61	0.01	0.41	0.0	0.62	0.01	0.47	0.01
mlp_model	DRFP	JNJ HTE+PMC 2024	Outter Clusters	0.45	0.0	0.42	0.0	0.47	0.0	0.65	0.01	0.39	0.0	0.58	0.01	0.4	0.0
knn_model	DRFP	JNJ HTE+PMC 2024	Outter Clusters	0.52	0.0	0.46	0.0	0.78	0.0	0.71	0.0	0.08	0.01	0.64	0.0	0.99	0.0
rf_model	DRFP	Dreher 2014	Outter Clusters	0.5	0.0	0.51	0.02	0.0	0.0	0.43	0.03	0.74	0.01	0.0	0.0	0.0	0.0
rf_model	DRFP	Dreher 2016	Outter Clusters	0.51	0.0	0.59	0.03	0.03	0.0	0.6	0.02	0.69	0.02	1.0	0.01	0.01	0.0
rf_model	DRFP	Dreher Doyle 2018	Outter Clusters	0.5	0.0	0.44	0.01	0.55	0.0	0.34	0.01	0.02	0.01	0.38	0.0	0.98	0.0
rf_model	DRFP	HiTEA 2024	Outter Clusters	0.5	0.0	0.59	0.02	0.0	0.0	0.57	0.03	0.66	0.04	0.97	0.09	0.0	0.0
rf_model	DRFP	AZ ELN 2023	Outter Clusters	0.49	0.03	0.47	0.02	0.53	0.06	0.43	0.01	0.37	0.07	0.45	0.03	0.67	0.14
rf_model	DRFP	Roche 2023	Outter Clusters	0.62	0.01	0.59	0.01	0.51	0.01	0.57	0.02	0.71	0.01	0.65	0.02	0.41	0.01
rf_model	DRFP	JNJ HTE 2024	Outter Clusters	0.5	0.0	0.6	0.02	0.0	0.0	0.59	0.02	0.66	0.03	0.1	0.3	0.0	0.0
rf_model	DRFP	JNJ HTE+PMC 2024	Outter Clusters	0.54	0.01	0.54	0.01	0.4	0.03	0.56	0.03	0.62	0.03	0.58	0.03	0.31	0.03